How to Fix an Inherited Report

A Toolbelt for Dissecting, Triaging, and Rebuilding the Report Nobody Wants to Touch

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Fun Fact: this font is dyslexic friendly! Ask me how to use it for free after the session.



About Me ~ Kristyna Ferris

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Blog

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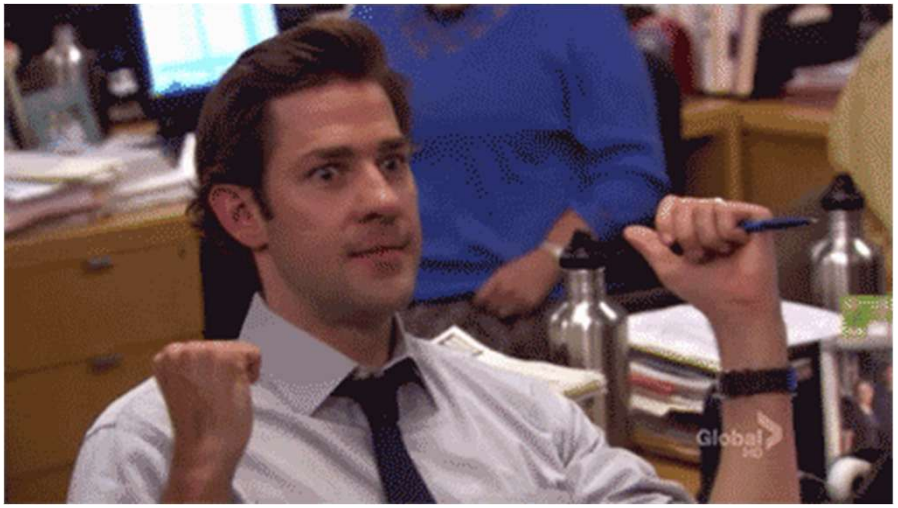
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The Big Break



“Hey, we have this report the previous VP used. It’s really slow, and the new VP wants it revamped to answer their questions.”

Great, you think. **I can knock this out, no problem!**

More facetime = more visibility = more opportunities.

Then You Open the Report...

50	100+	6
tables	measures	calculated tables

Zero documentation. The person who built it left two years ago.



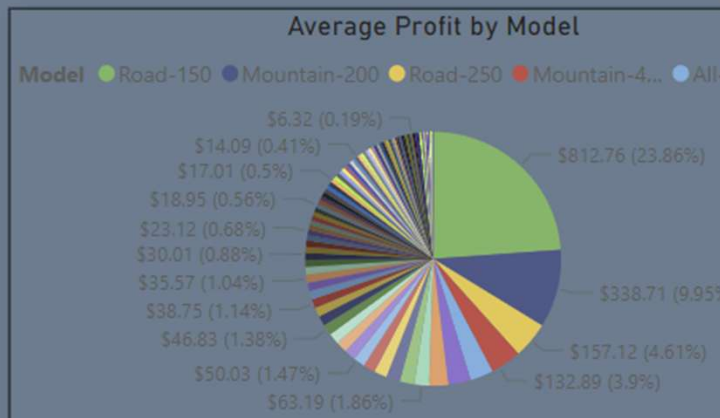
Terrible Report

(Blank)	FY2019	FY2021
FY2018	FY2020	

275K
Quantity Ordered

397
Product Count

Revenue by State-Province



Subcategory | Model | Product	Product Count	Quantity Ordered	Average
Road Frames	70	11,747	
Road Bikes	65	47,148	
Mountain Frames	40	11,620	
Mountain Bikes	38	28,321	$
Touring Bikes	22	14,751	
Touring Frames	18	3,725	
Jerseys	16	22,711	
Handlebars	14	3,950	
Wheels	14	5,263	
Tires and Tubes	11	18,006	
Gloves	9	12,983	
Helmets	9	19,541	
Saddles	9	2,145	
Pedals	7	3,931	
Shorts	7	9,965	
Socks	4	5,217	
Bib-Shorts	3	3,111	
Bottles and Cages	3	10,552	
Bottom Brackets	3	921	
Total	**397**	**274,776**	

Model	Product	Product Count	Cost
All-Purpose Bike Stand	All-Purpose Bike Stand	1	$14,807.03
Bike Wash	Bike Wash - Dissolver	1	$9,868.38
Cable Lock	Cable Lock	1	$11,199.38
Chain	Chain	1	$6,955.63
Classic Vest	Classic Vest, L	1	$4,916.04
Classic Vest	Classic Vest, M	1	$54,242.72
Classic Vest	Classic Vest, S	1	$100,862.00
Cycling Cap	AWC Logo Cap	3	$52,432.64
Fender Set - Mountain	Fender Set - Mountain	1	$17,435.68
Front Brakes	Front Brakes	1	$37,308.65
Total		**397**	**$97,257,907.95**

PANIC



Who made this hot pile of garbage?

And who's been using it?

How can I make this something beautiful in any sort of reasonable amount of time?

Will THIS be my legacy?

Take a Breath: Three Phases



PHASE 1

INSPECT

Know what you inherited before you touch it

- Save a safe copy
- Interview stakeholders
- Dissect the code with AI
- Share feedback with AI



PHASE 2

ARCHITECT

Design the target and grab the quick wins

- Quick-win checklist
- Visual best practices
- Model best practices



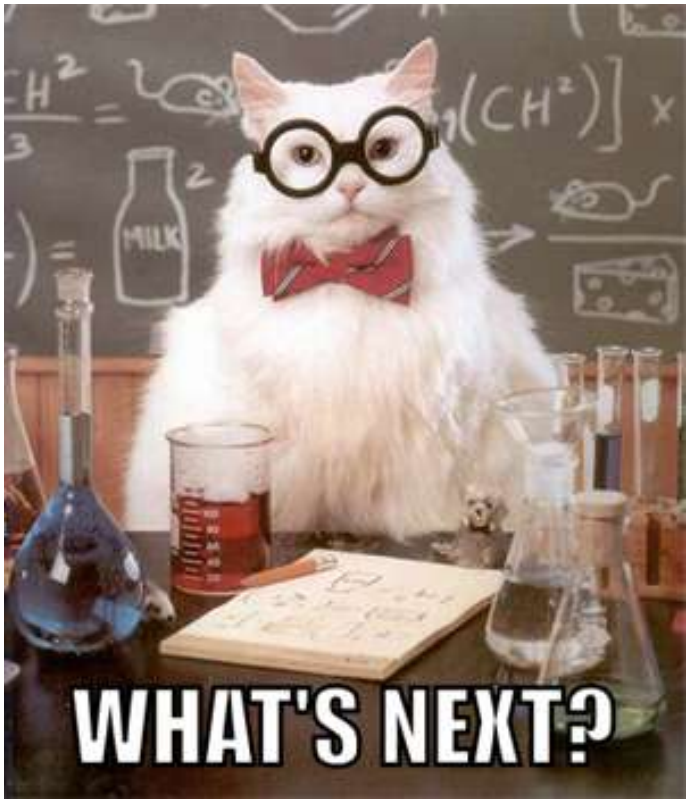
PHASE 3

BUILD

Rebuild it right, with AI as a helper

- AI rebuild guardrails
- Verify the numbers
- Tools for the job

Practical First Steps



INSPECT

1. **Save a copy** before changing anything.
2. **Meet with stakeholders** to learn what works and what doesn't. The goal is trust.
3. **Break it down** into manageable tasks.
4. **Define "done"** together to avoid a development spiral.
5. **Accept** that some things you cannot fix.

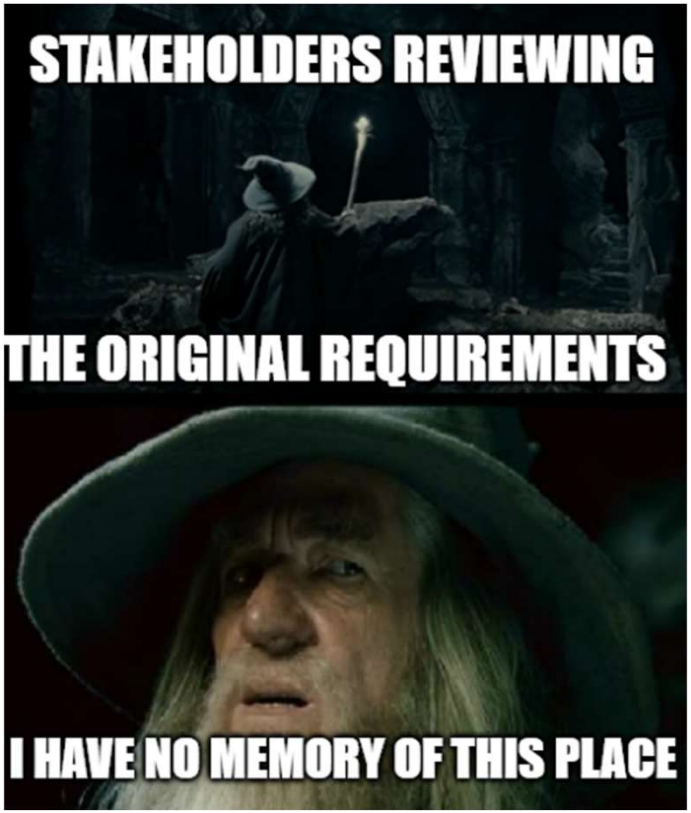
Meeting with the Stakeholders



Be nosy. Ask TONS of questions.

- What decisions will this data drive?
- What are the “make more money” metrics that matter most?
- What grain and scope does the data need?
- Who uses this report, and how do they slice it?
- What dimensions give context to the metrics?
- Where does the data live today?
- Are there data quality landmines waiting to explode?
- How often do these questions need answering?

Pro Tip: Hold Them Accountable



INSPECT

Write the answers down and email them to the stakeholders.

It creates a paper trail for when scope creep shows up uninvited.

Even better: put the answers on a **hidden page in the report itself.**

The purpose of the report travels with the report.

Dissect the Code with AI



Let AI read all fifty tables so you don’t have to. Then verify what it tells you.

1

Save as PBIP

Model and report become readable text files

→

2

Point AI at it

Open the project folder in Claude Code

→

3

Ask for an inventory

Tables, measures, calc tables, and where each is used

→

4

Verify

Cross-check with Measure Killer before you delete

Prompts I Actually Use





Inventory

List every table, measure, and calculated table, and which visuals use each one.



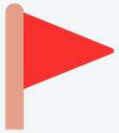
Dead weight

Which measures and columns aren’t used by any visual or other measure?



Translate

Explain this measure in plain English and flag anything that looks wrong.



Red flags

Find bidirectional and many-to-many relationships, auto date tables, and calculated tables.

AI reads the files. You own the answers.

What’s Working, What’s Not

INSPECT

Meeting transcript

Your notes and requirements

The PBIP code

KEEP

- Fiscal year filter
- Subcategory → Model → Product table
- Quantity Ordered

FIX

- “Profit Margin” is dollars, not %
- State funnel → revenue by territory
- Blank fiscal year

CUT

- 40-slice pie chart
- Scatter chart
- Cost table nobody can tie out

Prompt: *“Here’s the stakeholder transcript and the PBIP. Sort every visual and measure into keep, fix, or cut, and quote who said so.”*

Demo time!





- Save the report as a PBIP
- Dissect the code with AI
- Feed AI the stakeholder transcript
- Unused objects with Measure Killer

Quick Wins Checklist



Model

- [x] Remove unused tables and columns
- [x] Turn off Auto date/time
- [x] Move calculated tables to Power Query or upstream
- [x] Fix bidirectional and many-to-many relationships
- [x] Split high-cardinality datetime columns

Report

- [x] Run Performance Analyzer to find the slowest visuals
- [x] Cut each page to 6–10 insights
- [x] Swap pies and donuts for bars
- [x] Delete hidden and unused pages and visuals
- [x] Use explicit measures, not implicit aggregations

Visual Best Practices



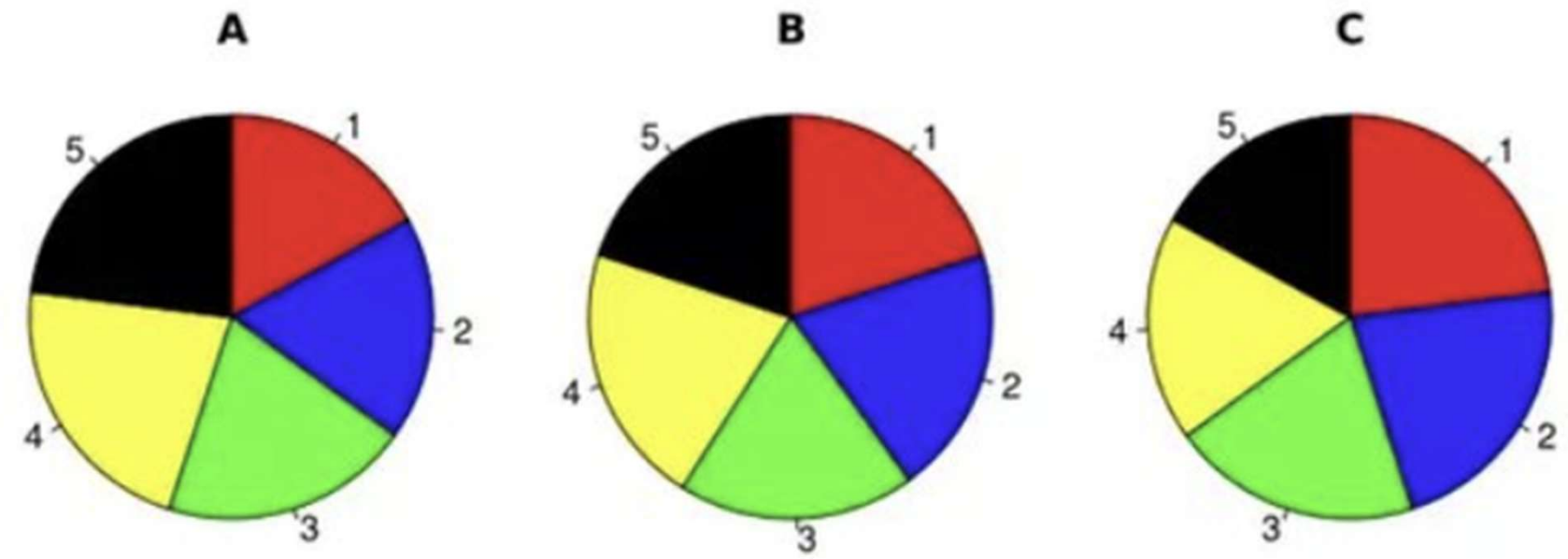
It’s all about TRUST

- About 6–10 insights per page
- Choose the right visual for the job
- Include trending visuals
- Group related metrics
- Size visuals to be readable
- Use a consistent color palette
- Add clear headings and labels
- Round numbers where it helps
- Lay it out to read top to bottom, left to right, like a newspaper

Reference: numerro.io, The Complete Guide to Designing Power BI Reports (link in notes)

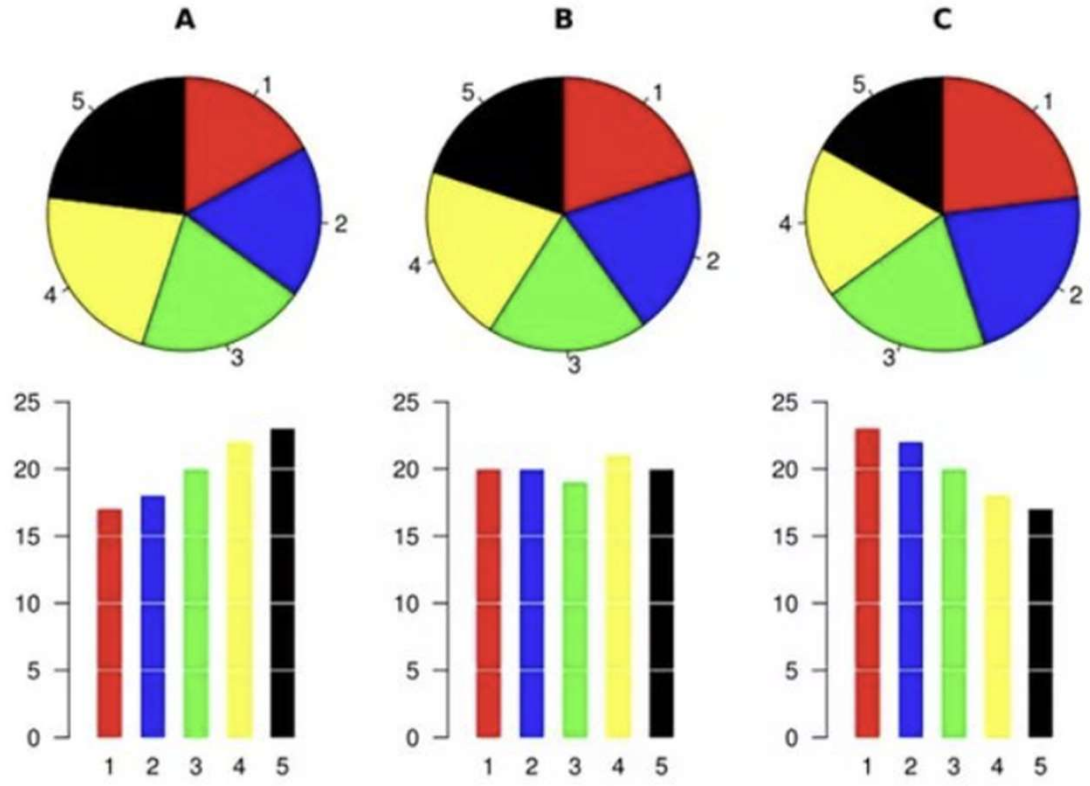
Rank These Slices, Biggest First

ARCHITECT



How Many Did You Get Right?

ARCHITECT



A 5, 4, 3, 2, 1

B 4, 5/2/1, 3

C 1, 2, 3, 4, 5

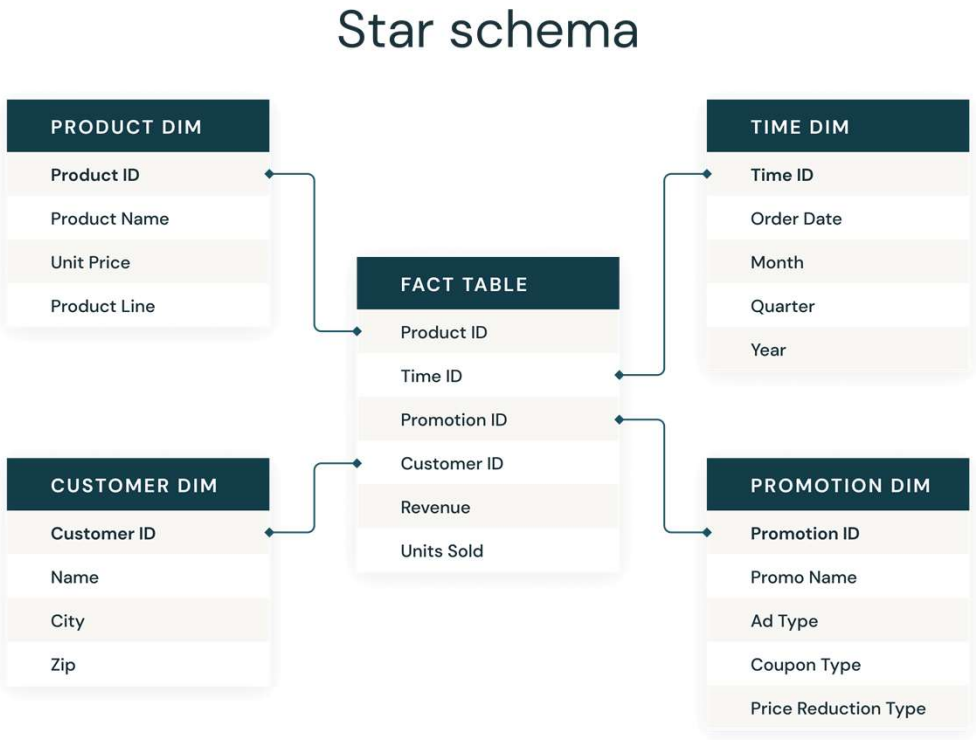
Model Best Practices



1. **Every column and table** feeds a visual or metric
2. **STAR SCHEMA ALL THE THINGS**
3. **No many-to-many** relationships (rare exceptions)
4. **Metrics are tested** and trusted by stakeholders
5. **Relationships** flow the right direction on the right fields
6. **Organize** the modeling view
7. **House metrics** in a measures table or folders
8. **Know when** to build from scratch

Star Schema, It’s Important

ARCHITECT



A proven design pattern for analytics.

Fact tables hold the numbers: sales, revenue, inventory.

Dimension tables give context: who, what, when, where.

Built for the VertiPaq engine: fewer tables, faster queries.

Unlocks slicers, hierarchies, and time intelligence out of the box.

Rebuild with AI, Safely



AI is a fast junior developer. ***Give it rules, small tasks, and a reviewer.***

1. **Work on a copy, never on live**

   PBIP in a git repo, so every change is a diff you can review

2. **Give AI your rules**

   A CLAUDE.md with your star schema, naming, and DAX standards

3. **Small batches**

   One table, measure group, or page at a time, then reopen in Desktop

4. **You approve every change**

   AI drafts. You decide what ships.

Trust, but Verify the Numbers



If the new report doesn't match the old one, nobody will trust either of them.



Snapshot the old

Run a DAX query of key measures by key dimensions against the original report



Rerun on the new

Same query, same filters, against the rebuilt model



Diff and explain

Every difference is either a fix you can explain or a bug

Tools for the Job





AI (Claude Code)

Reads PBIP files, inventories the model, and drafts changes to your standards



Measure Killer

Used vs. unused objects, lineage, and a best practice analysis



DAX Studio

See how queries execute in the engine and time them



Tabular Editor

Script model changes and run Best Practice Analyzer rules

All of these have free versions. Links are in the slide notes.

Helpful MCP Servers



Power BI Authoring

Microsoft · Local is GA

Edit models in Desktop, PBIP, or Fabric and run DAX. Has a read-only mode.

Fabric IQ

Microsoft · GA

Find reports and query models, read only. Security still applies.

Microsoft Learn

Microsoft · Free

Search the docs and code samples. No sign-in needed.

DAX Formatter

SQLBI · Free

Format and syntax-check the DAX that AI writes.

Tabular Editor 3

Tabular Editor · Paid

Agent edits the open model. Review or undo before saving.

Fabric MCP Server

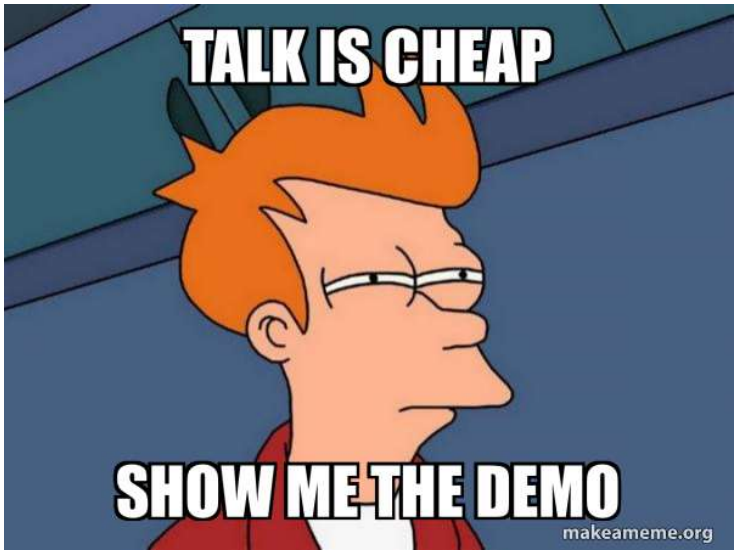
Microsoft · Open source

Fabric API docs, best practices, and OneLake files.

Start read-only. Let the agent look before you let it touch.

Demo time!





- Drop unused measures and calculated tables
- Reconnect visuals to the star schema
- Compare the numbers and the speed, before and after

Key Takeaways

1. Inspect: know what you inherited and why it exists before you touch it



2. Architect: grab the quick wins first; they buy you trust and time



3. Build: let AI do the reading and drafting; you own the decisions



4. Verify the numbers, or nobody will trust the rebuild



Don’t forget to connect, thanks for listening!

GitHub





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